
JPA FUNDING STRATEGIES

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Abstract

Abstract: We introduce an analysis and strategy to secure sustainable funding for the proposed Northeast California biomass supply management joint powers authority (JPA). These entities are proposed to create to improve forest product supply chain bottlenecks. An estimated \$1.5 mn is needed to start up and run the Northeast California JPA over three years. Startup funding for the JPAs for each of the pilots is being funded by the California Governor's Office for Land Use and Climate Innovation (LUCI). Long-term sustainable funding for each JPA will be challenging, especially in rural areas without the tax base to pay for wildfire mitigation authorities like Marin or JPAs that help make it easier to utilize the wood from thinning projects. A portfolio approach of securing public and private funding and securing self-sustaining revenue sources is recommended.

Keywords f, o, r, e, s, t, , h, a, l, ,, J, P, A, i, n, c

1 JPA Funding Strategies

1.1 Background

The large wildfires seasons of the past five years have been catastrophic for forests and communities. However, according to ?, they are not unprecedented in recent California history and are usually associated with periods of drought, e.g., 1920s and 2010s. researchers found that during the past century tree densities increased up to seven fold while average tree size decreased 50% in the Sierra Nevada (?). A chronic level of stress is created from high competition across trees leading stands to be less resilient to drought, disease, fire, and climate change. ? call for an aggressive approach to treat fire-suppressed stands with an ecological-based approach that reduces the total number of trees/acre, while maintaining stand heterogeneity.

1.2 Concerns

A significant cause for concern regarding the protection of old-growth forests in the United States is their drastic decline from historical levels due to logging and development, which threatens biodiversity and climate stability. ? highlight the fact that a large proportion of the remaining mature and old-growth forests on federal lands are still vulnerable to logging. The loss of these forests would not only diminish biodiversity but also release substantial amounts of stored carbon into the atmosphere. This underscores the urgent need to protect old growth as a natural climate solution.

Further compounding these concerns is the lack of a coordinated and effective national policy for old-growth conservation. ? point to a history of policy debates and unfinalized amendments, such as the National Old-Growth Amendment, which have failed to provide lasting protection. The authors argue for a more comprehensive approach that moves beyond simple harvest restrictions to include landscape-level planning, the establishment of reserves, the protection of climate and fire refugia, and the setting of specific goals for the recovery of species that depend on these unique habitats. This highlights the necessity of a long-term, ecosystem-based strategy to ensure the recruitment of new old-growth forests and the recovery of their associated biodiversity.

But let's be very clear, thinning dens stands of fire prone forests, is completely different from logging old growth. Furthermore, forest health treatments should not be one size fits all. Treating mixed conifer stands on the western slope of the Sierra Nevada requires different treatments than mesic forests in the Pacific Northwest. Managers should take steps to reduce fire risk, but still be required to consider negative impacts to biodiversity when planning any fire mitigation project.

1.3 Feedstock aggregation

Regardless of approach, thinned forests will create a lot of biomass and much of that biomass from forest health projects is ending up in burn piles or log decks without further utilization. Not only does this negate fire mitigation efforts, since burn piles and decks will promulgate or worsen fires, but the carbon from wood is at risk of not being sequestered.

Processing biomass from thinning is challenging. ? point out that supply guarantees, industry presence, transportation, and the value of the biomass are limiting to biomass utilization, whereas agency staffing, budgets, compliance, and partnership aggravated use rather than impeding progress. SIG recently completed a burn pile inventory across California showing massive tonnages of piles scattered through national forests and other lands. Yet burning those piles is likely more expensive than transporting them to a facility for processing (?).

Transportation subsidies to move this feedstock to central locations or nearby facilities could be a key missing component of the wildfire problem across the west.

1.4 JPA

Developing and creating the JPA is part of the Land Use and Climate Innovation (formerly OPR) funded California Forest Residual Aggregation for Market Enhancement (CAL FRAME) pilot in the state's northeast corner.

JPA finance strategy is adapted from the effort to secure sustainable funding for the proposed Northeast California biomass supply management joint powers authority (JPA) created to improve forest product supply chain bottlenecks. Developing and creating the JPA is part of the Office of Planning and Research (OPR) funded California Forest Residual Aggregation for Market Enhancement (CALFRAME) pilot. See the Introduction for more information on CAL FRAME and Chapter 2 for a summary of the CLERE Inc. [JPA Report](#) report to the Fall River Resource Conservation District.

What is a JPA?

Paraphrased from the Governments Working Together manual: Joint powers are exercised when the public officials of greater than two agencies agree to create another legal entity or establish a joint approach to work on a common problem, fund a project, or act as a representative body for a specific activity. Examples of areas where JPAs are used commonly include groundwater management, road construction, and habitat conservation.

A joint powers authority (JPA) is a separate government organization created by the member agencies, but is legally independent from them. A joint powers authority shares powers common to the member agencies, and those powers are outlined in the joint powers agreement ([Governments Working together, 2021](#)).

Partners developing the JPA are clear in the role of the organization acting as a critical wood utilization entity, but that it would not compete for funds with other organizations, especially RCDs or markets with private entities such as sawmills and licensed timber operators (LTOs). The approach outlined in this chapter takes a realistic, feasible revenue assessment appropriate for the region. It provides a hypothetical operating budget for running the JPA during the startup's first five years. Creating a stably funded JPA will be challenging with widely fluctuating changes in private and public funds.

2 Funding Options

The abundance of state and federal funding of the past five years has meant many agencies and organizations depend on grant funding for implementing restoration and infrastructure projects. Trump administration cuts in

2025 clearly demonstrate the vulnerability of relying too heavily on grant funding. Many nonprofits are familiar with the cyclic nature of many funding sources.

2.1 Traditional

The Northeast California CAL FRAME project recognized the need for grant funding to start the JPA, and realized the need for additional funding sources that can sustainably fund the JPA over time. These are described below and summarized in (Table 1).

1. **Endowment.** Create an endowment from the OPR funding tranche (plus other contributions) and invest it conservatively. Interest/gains from the principal would partially fund the JPA. We have assumed the seed funding for the JPA is \$2 mn from OPR with an additional \$500,000 from foundation, corporate, and individual contributions.
2. **Contributions.** Although the McConnell Foundation is an eventual, not immediate donor, casting a wider net at the state or Western Region level for foundations, corporations, and individuals that could give to the JPA is warranted. We suggest that rather than considering the contributions as grants to support the JPA, the Pilot Project Team should consider creating a campaign with a specific fundraising amount to start. This approach will make it easier to approach each donor with a specific amount to ask, make donors part of the overall campaign, and make smaller contributions from individuals meaningful and possible. We also recommend that initial contributions go to endowment creation.
3. **Federal and State Grants.** It is unlikely that the JPA will get operational funding from state and fed grants, but any proposals that it leads or participates in could charge overhead (~10%) plus directly bill salaries to cover some operating costs. Negotiating a higher indirect, or NICRA, to increase this value by a grant could be another long-term strategy to increase operating cost revenues.
4. **Member Contributions.** RCD and other feedstock aggregation members pay \$5k/yr to participate in the program. This would be unrestricted funding for the JPA.
5. **Fee for Service.** Western Shasta RCD, for example, works with nonindustrial forest owners and provides services for forest management plans with small forested LOs (The State funds the development of FMPs, but landowners cannot secure funds to implement the plans). JPA could create a similar service offering. However, it must ensure it does not compete with the region's RCDs.
6. **Sort yards.** Managing a sort yard for aggregated feedstock would be a strategic revenue source in line with the JPA mission. This would require substantial investment in equipment and a site to start. See the budget for a cost breakdown. A feasibility study would be helpful to understand total costs, potential revenue, and risk reduction associated with the yard.

2.2 Temporal Adoption

In general, the feasible options identified are traditional and similar to funding approaches for an RCD or nonprofit.

1. New finance options, such as environmental impact bonds, could be introduced over a longer time period and trialed at a small scale, then scale up when successful. These could also be other finance mechanisms, such as revolving loan funds (with low-interest loans), public-private partnerships, or bonds.
2. Other revenue-generating options could be tested in the same manner to ensure their effectiveness and introduce them to skeptical participants, financial managers, or government entities that are not used to working with alternative finance options.
3. The GANTT chart timeline includes testing, adopting, and implementing temporal finance mechanisms (Table 3).

Table 1: Analysis of funding types appropriate to aggregation JPAs. Timing roughly refers to the amount of time required to generate income. Difficulty is a qualitative scale from 1 = easiest to secure to 5 = most difficult.

Type	Pluses	Minuses	Timing	Difficulty
Endowment	Steady income from principal	Need significant amounts to generate income	May take a long time to build principal	5
Individual Contributions	Unrestricted funding	Much time to manage individual contributions	Instant	1
Public Grants	Access to large amounts of money	Reimbursable, administration costs can be high, stiff competition, cyclic	Public grants may take up to 1 year to contracting from proposal submittal	3
Private Grants	Often include unrestricted funding	Dependent on relationships to program officers, board members	Lengthy time for relationship building	2
Fee-for-Service	Steady income based on capacity and experience for services	Staffing, unevenness of demand	Long time to develop and market services	3
Sort Yard	Steady income, demand for wood high	Transportation costs, management, long-term supply	Permitting and capacity building may take a long time	4

2.3 Collaborative Finance

Collaborative finance is a conservation finance strategy that involves cooperative interaction between individual project developers, stakeholders, and finance providers. This process may or may not include traditional financial institutions (collaborativefinance.org). The term can be broadened to include finance developed by fair and equitable participation of stakeholders in a region, landscape, or watershed to address natural resource and infrastructure management needs, utilizing multiple forms of funding from public grants to private investment. Finance approaches may include outcomes-based finance models such as environmental impact bonds.¹

New Funding Options

New Finance Options Conservation finance options include environmental impact bonds like Blue Forest's Forest Resilience Bond, which leverages private investment from impact investors to support forest health projects, with repayment to investors over time supported by beneficiaries based on the savings and benefits from reduced wildfire risks and improved ecosystem services. A JPA could utilize environmental impact bonds on a trial basis at a small scale, then expand over an extended period. Identifying potential payor entities interested in paying for the avoided costs of wildfire with pro-active measures, such as thinning and prescribed fire, is worth pursuing immediately or as soon as the JPA hires an executive director.

We do not recommend solely relying on impact bonds since they take a lengthy time to develop and rely on a payor to provide avoided-cost funding. However, a bond could be a valuable complementary funding resource to other secured revenues. Other 'newer' finance options could include the following:

Carbon Markets Carbon markets offer an opportunity for the JPA, its members, and Implementing Entities to secure gap or stack funding for on-the-ground forest management activities, such as thinning, pruning, mastication, mechanical treatment, and even prescribed burning. Revenue realized through the carbon market

¹For a deeper discussion of collaborative finance approaches to financing water infrastructure and fire mitigation, see [Because It's Worth It](#) and [Conservation Finance: Towards a new model for landscape restoration](#).

could help the JPA to treat more forest land than otherwise possible and generate additional biomass that could be put under supply contracts to support existing and emerging infrastructure. Market prices for carbon credits vary depending on a given project's size and location, treatment type, and the carbon market or registry used. Carbon credits can be realized for projects of any size and located on federal, state, and private lands. See the Avoided Wildfire Emissions Protocol section as another approach that is easier to implement than carbon sequestration credits or markets since the avoided costs are already calculated, and proponents do not have to seek an entity to purchase the credits.

The National Forest Foundation implemented several voluntary carbon projects in California and the West following wildfires. Funded by corporate donors, the projects created carbon based on growth trajectories of planted trees. The credits were registered with the American Carbon Registry but immediately retired upon creation since trading carbon on public lands is disallowed.

Avoided Wildfire Emissions Spatial Informatics Group and Element Markets developed a forecast methodology under the Climate Forward program to recognize the climate benefits associated with fuel treatment activities that lower the risk of catastrophic forest fires and their emissions. Known as the Avoided Wildfire Emissions Forecast Methodology, the Climate Action Reserve protocol could provide complementary funding for thinning and prescribed fire projects to grants and private investments. The Protocol differs from carbon offsets in that forecasted mitigation units (FMUs) are issued for forecasted greenhouse gas reductions or removals. FMUs are used to mitigate anticipated future emissions, such as wildfires.

Embodied Carbon Although it may be several years from implementation, developing and marketing low-carbon building materials is another new opportunity. Buildings are a major source of greenhouse gas emissions, so building decarbonization is a California state priority. Embodied carbon is the lifecycle of greenhouse gas emissions from creating, transporting, and disposing of building materials (Carbon Leadership Forum, 2022). In other words, embodied carbon is any building's carbon footprint contained in its building materials. It differs from operational carbon, the carbon produced by a building's energy, heat, and lighting. The California Air Resources Board is developing a comprehensive strategy for embodied carbon and is seeking comments following the signing of ABs 2446 and 43 (CARB, 2024). More details on embodied carbon can be found in this slide deck from a recent California Air Resources Board workshop held on September 19, 2024.

Revolving Loan Funds Pooled funding sources such as impact bonds or revolving loan funds can help make a greater amount of funds available to more projects across a landscape. Typically offered at lower than market interest rates, revolving loan funds are self-replenishing pools of money utilizing principal and interest payments on existing loans to issue new loans. They have been used effectively on small to large scale to develop businesses, assist healthcare, and improve environmental outcomes. Revolving loan funds also provide much needed up front capital for project startup. They are flexible and can be used with more conventional funds such as grants and loans.

For example, through a coalition of public and private partners, the Southwest Wildfire Impact Fund intends to utilize resources from private investors and revenues from biomass generated from forest thinning to offset the financial burden for wildfire mitigation in the San Juan National Forest wildland-urban interface. The project fosters regional collaboration through shared project financing and implementation. It also creates the opportunity for scaling up forest treatments and fire reduction by creating a revolving loan fund that reinvests proceeds into additional projects, ensuring that capital is available for long-term re-treatment and expansion of forest health interventions.

Blue Forest is operating a revolving loan fund called the California Wildfire Innovation Fund that could expand into broader support of other forest health entities.

Parametric Insurance Parametric or index-based insurance covers the probability of a predefined event instead of indemnifying actual loss incurred (Swissre, 2018). These so-called trigger events are typically disaster (e.g., wildfire, flooding, hurricane, earthquake) related and measured through triggers such as wind speed, quake magnitude, acres burned/burn severity, or rainfall amount. Insurable triggers are modeled and must happen by chance. When the triggers are reached, a predetermined pay-out is made regardless of the

Table 2: Three-year hypothetical JPA budget showing revenue and expenses.

ITEM	YR1	YR2	YR3	TOTAL
REVENUE				
Contributions	60,000	60,000	60,000	180,000
Grants	60,000	120,000	150,000	330,000
Fee-for-Service	140,000	220,000	330,000	690,000
TOTAL REV-ENUE	260,000	400,000	540,000	1,200,000
EXPENSES				
Labor	340,000	370,000	410,000	1,120,000
Operations	20,000	30,000	40,000	90,000
Outreach	2,000	3,000	4,000	9,000
TOTAL EX-PENSES	362,000	403,000	454,000	1,219,000

sustained physical losses. Parametric insurance is meant to complement existing indemnity insurance but is increasingly used for post-disaster restoration funding in the natural world.

One of the earliest examples of parametric insurance use for nature recovery is the Mesoamerican Reef parametric insurance that provided \$800,000 for reef restoration following Hurricane Delta. The trigger was windspeed with a parameter greater than 100 knots. The funds came from the Coastal Zone Management Trust (Winters, 2020). Using insurance to protect forests and communities, likely through wildfire risk mitigation, could be a novel way to fund the JPA's activities throughout the region and provide complementary funds that could be particularly valuable post fire.

Tech-based Solutions Technology could help finance the JPA, connect payors to the JPA, and provide digital platforms that connect funders to implementers. Blockchain and digital solutions have mostly been applied to reforestation and carbon sequestration projects. It could be possible to simply set up a digital marketplace, where funders and implementers connect to implement forest health restoration and infrastructure projects. Developing a platform for aggregated funding could be a portion of a funded project or developed in tandem, rather than on its own.

3 Budget

CAL FRAME partners estimated an approximate annual budget averaging \$400,000 during the 1st three years of operation and based on similar operating expenses for RCDs in the region (Table 2).

3.1 Revenue

The revenue estimates are based on a startup endowment of \$1 million, regular indirect costs to cover operating from grants, and significant fee-for-service to forest landowners and possibly fees charged for use of wood from a sort yard. Ideally, the goal would be a stable revenue source to cover staff salaries and operational costs. It may be that a multi-year campaign is needed to get to a minimum level (\$2 million is suggested since that could generate ~\$100,000 annually as an endowment). We've made a conservative estimate for the start-up endowment, but the number may be higher but not substantially, e.g., \$2 mn.

Contributions include endowment, passive income, gifts, and member contributions. Contributions and gifts may come from local to regional foundations and corporations interested in the forest health world. Individual contributions always have the potential to add up to more than foundation and corporate gifts but take more time to manage. Creating a time-bound campaign with a specific fundraising goal replete with a thermometer to

show progress could be a great way to involve communities in the region through giving and creating outreach or communications opportunities at the same time to explain the need for the JPA and the importance of sustainable funding from the community to protect homes, infrastructure, and forests.

Revenue may also be derived from the feedstock supply insurance product being developed by members of the CALFRAME team and Ecostrat. Approximately half the startup funding could fund this mechanism or a parametric insurance model. These are in process, so more details will be provided once ready.

Grants in the budget will not compete with the RCDs. RCDs rely on various sources' grants to carry out their work, cover staff salaries, and keep the lights on at the office. Grant resources are limited, so a new entity such as the JPA cannot compete for those funds in any way.

On the other hand, the JPA may help administer a large grant across multiple RCDs to leverage more funds across a region. With the devolution of some state funding sources, this could be a great option to manage those funds and reduce the competition for funding resources since the grants solely go to local organizations. The fee-for-service section includes an item for grant administration. It also includes an option for implementing landowner forest plans since this may be a viable revenue source and may help source wood for the JPA. A sort yard that the JPA manages to source, centralize, and sell woody biomass is another revenue option but will require investment to be successful. Expenses for the sort yard are included in the expenses section.

JPA RCD competition

It is critical to incorporate into the JPA's bylaws and revenue-generating practices that grant fundraising does not compete with RCDs. It is possible a JPA may collaborate on a grant with one or more RCDs but should never submit grants for which an RCD qualifies.

Contingency funding should be written into the budget expenses. Adding 10-15% contingency line items to any secured grant would supplement that funding, although most grant funding contingencies are to be applied to implementation budget shortfalls. Other contingency funding sources could come from unrestricted funding (contributions/gifts) and a higher indirect rate.

3.2 Expenses

An annual expense budget of ~\$ \$400,000 is estimated for the JPA startup. The total expenses/year slowly ramp up each year of the budget with the idea that with additional secured revenue, the JPA would bring on more staff capacity and increase offerings, reach, or fee-for-service activities such as additional sort yards. The bulk of the expenses are for labor/staff, including an executive director, contracted feedstock manager, administrative bookkeeper, and various contracted services. Briefly, the staff responsibilities are the following:

- **Executive Director.** Manage board of directors, lead program development/outreach/education/fundraising/communications, develop an annual budget, manage contracts, recruit and manage staff.
- **Administrative Bookkeeper.** Track/invoice budget, develop cost allocation plan, manage grant reporting, administer payroll, manage AP/AR accounts, prepare 1099s and tax docs, develop grant budgets, and ensure financial compliance.
- **Feedstock Manager.** Source and sell feedstock from public and private lands, arrange transport to sort yard(s), manage sort yards, develop outreach materials, manage compliance documents for sort yard and aggregation, and maintain records.

Other expenses include operations and maintenance, audit/legal fees, insurance, equipment, software, travel, and bank fees. Expenses include communications, website, outreach and equipment, insurance, and land lease costs for a sort yard.

Table 3: Timeline for the JPA startup period.

Task	Sub-task	2025	2026	2027	2028	2029	2030
1. Revenue Generation	Endowment creation			X	X	X	
	Gift campaign	X	X	X			
	Fee-service			X	X	X	X
	Sort yard 1					X	
	Sort yard 2						X
2. Staffing	Executive director	X					
	Administrator	X					
	Feedstock manager		X				
3. Governance	JPA bylaws	X	X				
	LAFCO filing	X					

4 Timeline

An approximate timeline for key activities to fund and operationalize the JPA is shown in Table 3. Endowment seed funding and a capital campaign will be critical to initiate the endowment. Staffing is described under expenses and shows the approximate start time for each staff member. The development of the JPA bylaws will define governance, and the JPA will be created when the LAFCO filing is completed.

5 Recommendations

Expand explanation of these

1. Distinguish between start-up vs. operational funds.
2. Consider budget ramping
3. Tie expenses to realistic revenues
4. Consider temporal addition of non-traditional or higher risk revenue sources, but pilot them at a small scale first, then increase with success.
5. Fund a sort yard feasibility study.
6. Consider a feasibility study to determine revenue from realistic to moon-shot possibilities.
7. Examine insurance requirements and costs.
8. As revenue is secured, refine the budget and create a more detailed operating budget